Capstone Handbook

Master of Science in Software Engineering

Quantic School of Business and Technology

© 2025 Quantic Holdings, Inc. All rights reserved.

QUANTIC

Contents

Overview of the Capstone Project
Deliverables
- Assignment Requirements

The Capstone Journey
- Team Formation and Project Conception
- Sprints
- Working in Teams
- Sprint Planning Meetings
- Demonstration/Presentation Guidelines
- Quantic Submission Instructions
- Plagiarism Policy

Capstone Project Rubric
Capstone Presentation Rubric

Overview of the Capstone Project

The MS Software Engineering (MSSE) Capstone Project gives you the opportunity to integrate and apply your knowledge of software engineering languages, tools, methods, and project management in producing, as part of a team, either:

a. An **AI system** (a software system built incorporating the use of AI models and using AI engineering techniques) *or*
b. **Other software system**

Over the course of approximately three and a half months, you will draw on your software development or AI engineering expertise, apply appropriate engineering methodologies, and use collaborative tools learned throughout the program. This culminating experience simulates the role of a professional engineer and prepares you for real-world practice.

Over the course of the Capstone, you will continue to strengthen your skills in software project management, software architecture and design, software testing, AI-based applications and/or AI model fine-tuning and continuous integration/continuous delivery (CI/CD). You will apply the knowledge and techniques gained throughout your MSSE studies in a hands-on, project-based environment. In addition, you will develop essential teamwork skills by collaborating within a diverse, high-performing software engineering or AI engineering team. Because projects and their requirements vary, success will

© 2025 Quantic Holdings, Inc. All rights reserved.

require adaptability, a collaborative mindset, and a willingness to take responsibility for your own learning and contributions.

The experiential approach, commonly known as “learning by doing,” allows you to gain experience of working cooperatively and constructively as a part of a software or AI engineering team. The application of project management and software engineering skills to a valuable development task will allow you to practice and develop real-world software or AI engineering capabilities. Quantic faculty have extensive experience working in various industry sectors, and/or in consultancy roles and many have research experience in specific areas. Faculty will be a valuable resource to call upon if guidance is required to help you select and build a valuable software deliverable.

The software code you develop for your Capstone project will be hosted in a shared GitHub repository (with the option to make it public). This serves as a widely accepted and practical way to **showcase your engineering skills to potential employers and other professionals**. For this reason, you are encouraged to approach the project with diligence and a commitment to excellence - producing high-quality code and an outstanding deliverable **can provide significant, tangible benefits to your career after graduation**.

Learning Outcomes

When completed successfully, this project will enable you to:

- Work as part of an agile engineering team to analyze, design, develop, test and deploy a high-quality software system, or AI application or agent
- Utilize the appropriate engineering methodology and tools for your project to deliver a software system on-time
- Demonstrate a working deliverable that meets agreed requirements to product stakeholders
- Demonstrate the use of appropriate AI tooling to support software development
- Utilize appropriate software and AI engineering techniques and technologies throughout

Deliverables

The main deliverable of the MSSE Capstone is a working, high-quality **software system** or **AI system** meeting its user requirements. To deliver this software systemt you will be expected to have applied appropriate engineering methodologies and collaborative

© 2025 Quantic Holdings, Inc. All rights reserved.

tools (such as CI/CD tools), designed your product appropriately and utilized various technologies learnt during your studies.

While there are some requisite **deliverable components** (see below), there is no mandatory **format** for the final deliverable. You should include in your submission:

1) An (accessible) link to the **software repository** (a GitHub repo) containing the project's developed code. The code should be appropriately documented. You must share your repository with the GitHub account, **quantic-grader**.
2) The repository should include a link to the **deployed version** of the software or AI system (if a Web application).
3) The repository should include an (accessible) link to an agile **task board** (typically a Trello Scrum Board or other appropriate board) documenting all tasks carried out and features delivered
4) The repository should include a **design and testing document** listing and explaining 1) design and architecture decisions made, including any software and architectural patterns used, and 2) details of software testing implemented
5) A link to a recording of a **demonstration/presentation** of the working software or AI system at the conclusion of the Capstone Project. The deliverable if a Web application should be deployed to a free tier hosting environment such as Render, Railway etc. or any other of your choice.

Assignment Requirements

- This is a **group assignment** with groups of up to six (6) members, but you also have the option to **work individually** if you prefer. Group assignments are student-led, but the engagement team will offer support in the selection of an appropriate team.
- In order to receive full credit for the project, **all** elements **must** be completed.
- Your final Capstone deliverables should be submitted electronically at the end of the program. This will include a recorded presentation/demonstration of your working software or AI system. See Submissions and Grading below for more information.

The Capstone Journey

Now that you have this handbook, you are ready to start your MSSE Capstone journey! You can think of the Capstone journey as taking place over three phases: **team formation and project conception,** a series of **agile development sprints**, and the final **submission** phase.

© 2025 Quantic Holdings, Inc. All rights reserved.

Team Formation and Project Conception

Your first task on your Capstone journey is to assemble into a team of a maximum of 6 group members, ready to take on an engineering project! You will be able to select your team members. Look out for the signup sheet on Slack.

Once assembled, teams will have the opportunity to conceive of and select their own project topic based on its relevance to group member's interests/work experience or the opportunity it provides to practice technology skills of interest to the group members. It is recommended that potential project topics could be surfaced for consideration, based on the following:

- Any team member can suggest a software application or AI use case from their current workplace or past work experience. Deliverables based on real-world considerations and constraints support a rich learning experience. If undertaken, simulated data can be used.
- In identifying potentially valuable AI use cases and solutions you are encouraged to consider using approaches covered in the AI for Business specialization.
- A hypothetical or novel application or product can be suggested
- Projects that apply AI engineering techniques in a novel way to real-world use cases are encouraged—but not required.

All team members will then need to reach agreement on the specific project topic selected to be developed.

Once your project topic has been finalized, a group member will then act as Product Owner for the software or AI system project. If the project topic relates to a real-world workplace need known to a particular group member, then that group member would be the Product Owner.

The Product Owner will need to develop an initial list of user stories for the software or AI system.

Sprints

Once you have conceived and selected a project, you will be expected to arrange a start-up meeting and decide on the upcoming sprint timeline and carry out your first sprint planning meeting. The project team must complete **at least three sprints** by the end of the Capstone journey.

In the first meeting:

© 2025 Quantic Holdings, Inc. All rights reserved.

- A Scrum Master will be nominated and agreed upon by the team
- The Product Owner will discuss the scope of the project and the initial user stories with the team
- A prioritized product backlog will be discussed and developed by the team
    - Product backlog items could consist of user stories, features, requirements or other tasks
- The team will decide on the collaborative tools and software technologies to be used during the project for software development
- A sprint timeline will be agreed upon including the end date of the first sprint
- The first sprint planning meeting:
    - The first half of the sprint planning meeting will establish which of the user stories or other items from the product backlog will be delivered by the end of the first sprint—the sprint backlog
    - In the second half of the sprint planning meeting, the team members along with the Scrum Master, determine the individual tasks to be carried out and by whom to implement those agreed user stories or tasks during the sprint

The majority of the time in your Capstone journey will be spent carrying out several agile software development **sprints**. The team will:

- Carry out **at least three sprints** during the Capstone journey
- Maintain a (Web-based) Scrum Board documenting the product backlog and the user stories being implemented in each sprint and their specific constituent tasks and the completion status for each user story and task
- During each sprint, the team will utilize CI/CD practices and tools, to collaboratively update, test and develop the shared code base that will be stored in their project's GitHub repository
- At the end of each sprint the team will provide a recording of a software demonstration of their working software to-date to the Product Owner that will be used for sprint review
- Hold a sprint planning meeting with the team at the start of each new sprint and repeat for the next sprint cycle

If the software or AI system is a Web application, by the time of submission, it should be deployed to a free tier production hosting environment such as Render, Railway etc. or any other of your choice.

© 2025 Quantic Holdings, Inc. All rights reserved.

Working in Teams

In each team, there will be different roles, including Product Owner, Scrum Master and Team Members including one or more Code Owners (approve pull requests to the GitHub repo). The Product Owner and Scrum Master will contribute equally to software development work as the other team members. A person may well have a particular skill suited to a particular element of the project work and the team should aim to optimally use and assign tasks taking this into account. You will develop your understanding and use of team dynamics and learn how best to handle conflicts and resolve differences.

Your team should meet in regular time-boxed scrum meetings to discuss progress on your Capstone around **once a week**. You should also attend, and meaningfully participate in, as many webinars and facilitated events offered as part of the Capstone journey as you can.

Each event is an opportunity for you to further develop your skills, apply your knowledge and learning, and to meet—and learn from—others.

Sprint Planning Meetings

Sprint planning meetings, held at the start of each sprint, must be attended by all team members to plan for the coming sprint. In about a week, there will be a designated Check-in Week where you can schedule a **check-in meeting** with a Quantic faculty member to ask questions and receive feedback on your work in progress.

Reporting and Submission

Below are the elements that **must** be provided in order to create a comprehensive Capstone project submission and receive **full credit** for your submission. This list provides guidance but the format is not mandatory. You should follow a format that is appropriate for your project and solution. There is no minimum or maximum length to the submission.

Required contents for your submission:

- A **link** to an accessible **software repository** (a GitHub repo) containing all your developed code. The code should be appropriately documented. You must share your repository with the GitHub account, quantic-grader.
    - The GitHub repository should include a link to the **deployed version** of the software or AI system (if a Web application).

© 2025 Quantic Holdings, Inc. All rights reserved.

- The GitHub repository must include a **link** to an accessible **agile task board** (typically a Trello Scrum Board or other appropriate board) documenting all tasks carried out by group members, and user stories delivered
- The GitHub repository must also include a detailed **design and testing document** listing and explaining:
  - i) design and architecture decisions made. This should include amongst other items, technologies and architectural choices made and reasons for these. This should include any software and architectural patterns used and reasons used. It should also include what deployment options are recommended for the software (e.g. on-premises or cloud) including relative cost implications of the choice, and;
  - ii) software testing carried out. This should document all testing done including any and all automated tests used and reasons why.

- A link to a recorded **demonstration/presentation** of the working software or AI system, involving screen capture of the software being used with voice over—all group members must speak and be present on camera. The presentation should be between 15 and 20 minutes in length.

Demonstration/Presentation Guidelines

- The demonstration/presentation should offer a summary of the software or AI system solution developed, and a demonstration showing all its functional capabilities, with the demonstrators showing the working, deployed deliverable via screen share.
- The presentation should be submitted via a recording where all members are present and visible on-screen AND your demonstration is recorded on your screen.
- In your presentation, you <u>must</u>:
  - <u>Present a government-issued ID to the camera so your name and picture are clearly visible and legible.</u>

Quantic Submission Instructions

Submit your final project deliverable electronically as (accessible) links to each of the required deliverable components using the “Submit Project” buttons on your Quantic dashboard. In order to record your presentation, we recommend recording with Zoom and submitting your work either as a link to an **unlisted** Youtube video or to a file uploaded to a stable storage system (Google Drive, OneDrive, Dropbox, etc.) Please do not use WeTransfer or another website with expiring links, as sometimes the link will

© 2025 Quantic Holdings, Inc. All rights reserved.

expire before submissions are processed for grading. For specific recommendations on recording options, check out this support article.

Plagiarism Policy

Quantic takes academic integrity very seriously—we define plagiarism as: “Knowingly representing the work of others as one’s own, engaging in any acts of plagiarism, or referencing the works of others without appropriate citation.” This includes both misusing or not using proper citations for the works referenced including any software code of others, and submitting someone else’s work as your own. Quantic monitors all submissions for instances of plagiarism and all plagiarism, even unintentional, is considered a conduct violation. It is important to be conscientious when citing your sources. When in doubt, cite! You can also find more about our plagiarism policy here. Please see guidance on appropriate citation of other programmer’s code if and where used that should be followed here.

Capstone Journey Checklist

Find a team to work with	▢
Conceive of potential projects and agree on and select a specific project. Assign a Product Owner and Scrum Master.	▢
Meet with your team for an initial scoping and first sprint planning meeting	▢
During the sprint, your team should meet weekly and implement the agreed user stories, tasks or requirements	▢
Submit a recording of a demonstration of your software or AI system to your Product Owner by the sprint end date	▢
(iterate) For each subsequent sprint: - Hold a sprint planning meeting - Implement planned user stories, tasks or requirements during the sprint - Submit a recorded software demo to your Product Owner by the sprint end date	▢
Attend webinars and events (optional)	▢
Submit your final project deliverable including final recorded software demonstration presentation to Quantic	▢

© 2025 Quantic Holdings, Inc. All rights reserved.

Capstone Project Rubric

Scores 3 and above are considered passing. Students who receive a 0-2 will not receive credit for the submission and must revise and resubmit to receive a passing grade.

Score	Description

5	• Addresses ALL of the Capstone requirements including, but not limited to:
	  ○ Provides a software repository showing all developed project code. All code should be appropriately documented.
	  ○ Provides a link to the deployed version of the software or AI system (if a Web application).
	  ○ Provides an up-to-date task board showing completion of all agreed user stories and tasks
	  ○ Evidences well-designed and well-tested code via a detailed design and testing document. This should include any software and architecture patterns used and reasons for their use. It should also detail all testing done and testing methods used.
	  ○ Appropriate software engineering methodology and collaborative software engineering tools, including CI/CD tools, have been used
	• An outstanding recorded software demonstration that clearly shows an outstanding software system.
	• Shows initiative demonstrated by the project software deliverable going above and beyond the minimum Capstone requirements

4	• Addresses MOST of the Capstone requirements including, but not limited to:
	  ○ Provides a software repository showing all developed project code. All code should be appropriately documented.
	  ○ Provides a link to the deployed version of the software or AI system (if a Web application).
	  ○ Provides an up-to-date task board showing completion of most agreed user stories and tasks
	  ○ Evidences fairly well-designed and fairly well-tested code via a detailed design and testing document. This should include any software and architecture patterns used and reasons for their use. It should also detail all testing done and testing methods used.
	  ○ Appropriate software engineering methodology and collaborative software engineering tools, including CI/CD tools, have been used

© 2025 Quantic Holdings, Inc. All rights reserved.

	• An excellent recorded software demonstration that clearly shows an excellent software system. • Shows good initiative

3	• Addresses SOME of the Capstone requirements including but not limited to:
	  ○ Provides a software repository showing all developed project code. Most code is appropriately documented.
	  ○ Provides a link to the deployed version of the software or AI system (if a Web application).
	  ○ Provides an up-to-date task board showing completion of some of the agreed user stories and tasks
	  ○ Evidences moderately well-designed and moderately well-tested code via a detailed design and testing document. The document should include any software and architecture patterns used and reasons for their use. It should also detail all testing done and testing methods used.
	  ○ Appropriate software engineering methodology and collaborative software engineering tools have been used
	• A very good recorded software demonstration that clearly shows a very good software system.
	• Shows some initiative

2	• Addresses A FEW of the Capstone requirements including, but not limited to:
	  ○ Provides a software repository showing all developed project code. Little of the code is appropriately documented.
	  ○ Does not provide a link to the deployed version of the software or AI system (if a Web application).
	  ○ Provides an up-to-date task board showing completion of a few agreed user stories and tasks
	  ○ Evidences limited design and testing of code via the design and testing document. Provides limited evidence of any software and architecture patterns used and reasons for their use. It provides limited information on testing done and testing methods used.
	  ○ Appropriate software engineering methodology and collaborative software engineering tools have been used, but with very little use of appropriate CI/CD tools.
	• A basic recorded software demonstration that shows a good software system.
	• Shows little initiative to go above and beyond the minimum requirements

© 2025 Quantic Holdings, Inc. All rights reserved.

1	• Addresses the Capstone project requirements but MOST components are missing including, but not limited to:
	  ○ Provides a software repository showing all but limited developed project code. The code is not well documented.
	  ○ Does not provide a link to the deployed version of the software or AI system (if a Web application).
	  ○ Provides an up-to-date task board showing completion of very few agreed user stories and tasks
	  ○ Evidences poorly designed and tested code via the design and testing document. Provides no evidence of any software and architecture patterns used. It provides no information on testing done and testing methods used.
	  ○ Appropriate software engineering methodology and collaborative software engineering tools have largely not been used, and no use of CI/CD tools made
	• No recorded software demonstration is provided.
	• Shows no initiative

0	• The assignment is significantly incomplete, plagiarized, and/or most vital components are missing

Capstone Presentation Rubric

Scores 2 and above are considered passing. Students who receive a 1 or 0 will not receive credit for the Capstone submission and must revise and resubmit to receive a passing grade.

Score	Description
5	• Provides a thorough, clear and concise demonstration of the user stories implemented by the Capstone project software artifact or AI system for a range of user inputs • The deliverable operates correctly throughout the demonstration • The presentation is professional and the screen share is clear and legible • The students are clearly visible and audible during the presentation • The presentation is between 15 and 20 minutes long

© 2025 Quantic Holdings, Inc. All rights reserved.

4	• Provides a thorough, clear and concise demonstration of the user stories implemented by the Capstone project software artifact or AI systems • The deliverable operates correctly throughout the demonstration with only minor exceptions • The presentation is professional and the screen share is clear and legible • The students are clearly visible and audible during the presentation • The presentation is marginally outside of the time requirement.
3	• Provides a clear and concise demonstration of the majority of user stories implemented by the Capstone project software artifact or AI system • The deliverable operates correctly throughout the demonstration with only a few exceptions • The presentation is professional and the screen share is clear and legible • The students are clearly visible and audible during the presentation • The presentation is significantly outside of the time requirement.
2	• Provides a mainly clear demonstration of some of the user stories implemented by the Capstone project software artifact or AI system • The deliverable operates correctly throughout the demonstration with a number of exceptions • The presentation is generally professional and the screen share is clear and legible • The students are clearly visible and audible during the presentation with some exceptions • The presentation is significantly outside of the time requirement
1	• Provides a very limited demonstration of some of the user stories implemented by the Capstone project software artifact or AI system • The deliverable encounters significant issues/ does not run correctly during the demonstration • The presentation is unprofessional and the screen share is not clear and legible • The students are not visible and/or audio is missing during the presentation • The presentation is significantly outside of the time requirement
0	• The assignment is incomplete, plagiarized, and/or is missing most vital components.

© 2025 Quantic Holdings, Inc. All rights reserved.